higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 2

(6050342)

18 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Process flow diagrams are also called …

A pipe specifications.
B level controllers.
C flow sheets.
D major bypasses.

1.2 … are used to relay extra information that is not described by the line structure or equipment symbol.

A Legends
B Labels
C Valves
D Fittings

1.3 One of the clearest ways of presenting pipe routing in a process plant is with the assistance of …

A isometric drawings.
B lines.
C symbols.
D process flow diagrams.

1.4 … has poor resistance to acids and alkalis.

A Aluminium
B Copper
C Steel
D Polymers

1.5 Colour coding in pipes helps to …

A predict when the pipe will burst.
B protect against corrosion.
C protect pipe contents against frost.
D identify what a pipe contains.

(5 × 1) **[5]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 Process flow diagrams do not indicate flanges.

2.2 AutoCAD is not used to draw flow sheets.

2.3 Steel has strong resistance against corrosion.

2.4 Mechanical steam traps work due to the difference in density between the steam and the condensate.

2.5 When the valve is fully opened, the disc is retracted into the area known as the bonnet.

(5 × 1) **[5]**

QUESTION 3

Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (3.1–3.5) in the ANSWER BOOK.

3.1 The water that forms part of our rivers and lakes is called … water.

3.2 Wastewater contains a lot of harmful …

3.3 Vibrating screens use … to move screens and thereby separate the particles.

3.4 The discs of screens have a … shape with a number of teeth protruding from them.

3.5 Two screens are commonly referred to as …

(5 × 1) **[5]**

TOTAL SECTION A: 15

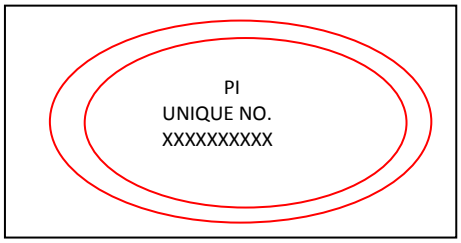
SECTION B

QUESTION 4

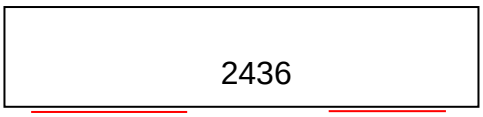
4.1 List THREE purposes of process flow diagrams. (3)

4.2 Give the descriptions of the symbols below:

4.2.1



4.2.2



(2 × 1) (2)

4.3 Write brief notes on the pH scale. (5)

4.4 Explain how an insulation material can be protected from damage. (2)

4.5 Name THREE things that can be used to manually operate a valve. (3)

[15]

QUESTION 5

5.1 List SIX components of a single-stage centrifugal pump with only one impeller. (6)

5.2 Reciprocating pumps are also known as positive displacement pumps.

5.2.1 Name TWO types of reciprocating pumps. (2)

5.2.2 Explain how the pumps in QUESTION 5.2.1 transfer fluids. (6)

5.3 Define *wet gas.* (2)

5.4 A bicycle pump is an example of a simple reciprocating compressor.

Explain the downward and upward motion of the piston of the bicycle pump. (4)

[20]

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QUESTION 6

6.1 Describe the TWO purposes of storing materials. (2)

6.2 Explain the term *bulk storage.* (2)

6.3 List FIVE typical bulk storage methods used in industries. (5)

6.4 Name FOUR categories of materials to be stored. (4)

6.5 Define *storage tanks.* (2)

6.6 Companies in the processing industries make use of steel, plastic and fibre drums to store and ship a variety of chemicals, catalysts and hazardous materials.

6.6.1 Why is it important for the manufacturers to adhere to strict production and shipment standards? (2)

6.6.2 Why should drums be stored indoors? (2)

6.6.3 What will happen to drums if stored outside? (1)

[20]

QUESTION 7

7.1 Define the following terms:

7.1.1 Filtrate

7.1.2 Concentrate

7.1.3 Turbidity

(2 × 3) (6)

7.2 Draw a freehand diagram to illustrate the filtration mechanism. (5)

7.3 Explain the operation of a slow sand filter. (8)

7.4 With the aid of a diagram, explain the working principles of a rectangular thickener or settling tank. (6)

7.5 What is the main purpose of a reverse-flow cyclone? (2)

7.6 List THREE types of crystallisation equipment used in industry. (3)

[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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